William Kwako

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EDUCATION

M.S., Computer Science – Interactive Intelligence, Georgia Tech, GPA: 4.0 — August 2023 – August 2026
B.A., Physics, Willamette University — August 2015 – May 2019

TECHNICAL SKILLS

- **Languages**: Python, C#, Java, JavaScript, SQL, M/MUMPS
- **AI & ML:** NumPy, Matplotlib, pandas, Stable-Baselines3, Gymnasium, PyTorch, scikit-learn, OpenAI API, Anthropic API, Prompt Engineering, LangGraph, Chroma, RAG, LLM Evaluation
- **Frameworks & Tools:** Django, FastAPI, Git, Microsoft Azure, Linux, nginx, GitHub Actions (CI/CD), Claude Code, LaTeX
- **Healthcare:** Epic, revenue cycle, X12 EDI (837I/837P, 835), HIPAA/PHI, FHIR, Medicare coverage policies (LCD/NCD)

PROFESSIONAL EXPERIENCE

Technical Solutions Engineer – Epic Systems Corporation — December 2019 – December 2022

- Owned claims and remittance support for five health systems, managing up to 100 open issues at a time and resolving roughly 10-20 per week
- Troubleshot production claims issues live with revenue cycle analysts on 10-20 client calls per week
- Implemented new payer requirements in claims and remittance configuration (837I/837P, 835), including claim attachments, PLBs, and state reporting
- Wrote and ran M/MUMPS queries against production databases containing PHI to diagnose client issues
- Led the Debugging Task Force; trained new hires on internal debugging tools, revised the onboarding curriculum, and wrote troubleshooting guides

Intern – Willamette University, Physics Department — May 2018 – July 2018

- Implemented 3D wavelet convolutions via FFT in Python to search for baryon acoustic oscillation signals in the 2MASS Redshift Survey; presented at AAS 233

PROJECTS AND RESEARCH

- **Predicting Medicare Coverage from FHIR Records with RAG and LLMs**
  Healthcare AI Research, 2026 | Website | Write-up & Code
  - Built a prior authorization checker that retrieves governing Medicare coverage policies using RAG, judging each clinical criterion in a FHIR record as met, not met, or insufficient information
  - Reached 98-99.5% policy selection and 89-94% decision accuracy on 72 synthetic cases (development and held-out policies), with under 3% false “covered” decisions
  - Deployed to an Azure VM (nginx, gunicorn, systemd, HTTPS) with CI/CD via GitHub Actions and SQLite request logging
  - **Tools**: Python, FastAPI, Chroma, Anthropic API, FHIR, Azure, GitHub Actions, SQLite

- **Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems**
  Educational Technology Research, Presented at NCME 2026 | Website | Write-up & Code
  - Built a web application generating personalized coding exercises to improve code comprehension
  - Designed and implemented a multi-model pipeline with conditional routing, prompt chaining, guardrails, and feedback loops across OpenAI and Anthropic APIs
  - Built a validation and repair loop that executes generated code, verifies it against user specifications via AST inspection, and escalates to a stronger model on failure
  - **Tools**: Python, Django, JavaScript, HTML/CSS, OpenAI and Anthropic APIs
- **Improving Users' Academic Writing Ability Through LLM-Generated Feedback**
  Educational Technology Research, 2026 | Write-up & Code
  - Built a Django web application that generates writing feedback by decomposing analysis across six specialized LLM agents using LangGraph, with a synthesis step reconciling critiques
  - Designed a controlled evaluation (order-swapped, cross-family judge): the graph beat a single same-tier call in all three matched experiments (16-6-8 over 30 comparisons)
  - Isolated analyzer strength as the bottleneck: upgrading the analyzers flipped results against a strong single call from 2-6 to 6-3, while upgrading only the synthesizer did not (3-6 to 2-6)
  - Ablated RAG (Chroma; genre exemplars, rubric criteria): no net benefit at n=10; reported as a null result
  - **Tools**: Python, Django, LangGraph, Chroma, HTML/CSS, OpenAI and Anthropic APIs
- **Benchmarking Reinforcement and Imitation Learning on a Deterministic Platformer: Jump King**
  Reinforcement Learning Research, 2026 | Write-up & Code | Paper
  - Built a full pipeline interfacing with a live commercial game via a custom C# mod (real-time state extraction), TCP socket communication, and a custom Gymnasium environment
  - Compared PPO, behavioral cloning (BC), and BC+RL per level: BC+RL completed all 43 levels (78-100% completion rate), while pure RL scored 0% on hazard and precision levels
  - Chained per-level agents reliably complete the full base game in ~10 minutes
  - **Tools**: Python, C#, Stable-Baselines3, Gymnasium, dnSpy, .NET

POSTERS AND PRESENTATIONS

- *Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems*
  Poster Presentation – National Council on Measurement in Education (April 2026)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*
  Oral Presentation – Willamette-Linfield intercollegiate symposium (May 2019)
  Poster Presentations – 233rd Meeting of the American Astronomical Society (January 2019), Murdock College Science Research Conference (November 2018)

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900)** October 2026